

Ayush Sharma Kaundinya

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GitHub: [icaruss](https://github.com/icaruss)

RESEARCH OBJECTIVE

Computer Engineering graduate with research experience in Artificial Intelligence, particularly in Computer Vision, Natural Language Processing, and Data Science. Research work includes YOLO-based object detection and transformer-based abstractive text summarization, alongside practical experience developing AI systems. Interested in pursuing graduate research in Artificial Intelligence, focusing on computational and data-driven methods for advancing intelligent systems.

EDUCATION

- **Gandaki College of Engineering and Science, Pokhara University** July 2021 – August 2025
Bachelor of Engineering in Computer Engineering Nepal
 - Relevant coursework: Artificial Intelligence, Natural Language Processing, Database Management Systems, Data Structures & Algorithms, Theory of Computation, Operating Systems, Programming Technologies, Cloud Computing, Data Mining, Image Processing and Pattern Recognition

PUBLICATIONS

J=JOURNAL, C=CONFERENCE

- [J.1] A. S. Kaundinya, P. Adhikari, M. B. KC, and P. Shrestha, “**An In-Browser Proctoring System Using YOLO-Based Object Detection and Gaze Analysis**,” *Journal of Artificial Intelligence and Capsule Networks*, vol. 7, no. 4, pp. 388–412, December 2025. DOI: [10.36548/jaicn.2025.4.005](https://doi.org/10.36548/jaicn.2025.4.005)
- [C.1] M. B. KC, A. S. Kaundinya, P. Adhikari, and S. Adhikari, “**Abstractive Summarization of News Articles Using BART**,” presented at The 1st International Conference on Emerging Trends in Engineering and Technology (ICET-2025). Published in *Pokhara Engineering College Journal*, vol. 3, no. 1, pp. 132–148, 2026. DOI: [10.3126/pecj.v3i1.93538](https://doi.org/10.3126/pecj.v3i1.93538)

RESEARCH AND TECHNICAL PROJECTS

- **In-Browser Proctoring System** [Code]
Tools: Python, YOLOv8, ONNX Runtime, DeepFace, JavaScript
 - Built a browser-based exam monitoring system combining facial recognition, gaze tracking, and real-time object detection to flag suspicious behavior during online exams, running at 25+ FPS in-browser
 - Fine-tuned YOLOv8s on a custom dataset and exported the model to ONNX format, enabling GPU-free, cross-platform inference without a server dependency
 - Research published in the *Journal of Artificial Intelligence and Capsule Networks* (see Publications, [J.1])
- **News Summarization Using BART** [Code]
Tools: Python, PyTorch, HuggingFace Transformers, vLLM
 - Fine-tuned the BART transformer on the XSum and CNN/DailyMail datasets (300k+ articles) for abstractive news summarization, achieving strong ROUGE scores on both benchmarks
 - Deployed the fine-tuned model using vLLM for high-throughput batched inference with low serving latency
 - Presented at ICET-2025 (see Publications, [C.1])
- **IntellyQ — Multi-Tenant AI RAG Backend** [Code]
Tools: Python, FastAPI, ChromaDB, PostgreSQL, Redis, OpenRouter, Clerk, SentenceTransformers
 - Developed a production-ready multi-tenant RAG backend enabling users to upload PDFs and query their documents through a multi-layer guardrail pipeline (rule filters, LLM classifier, decision engine)
 - Implemented hybrid BM25 and dense vector search with cross-encoder reranking, and SSE-streamed LLM responses via OpenRouter, supporting Solo and Enterprise account tiers
 - Built with Clerk authentication, PostgreSQL and ChromaDB persistence, Redis response caching with a circuit breaker, and async-safe SentenceTransformer embeddings; covered by 191 automated tests
- **Facial Emotion Detection** [Code]
Tools: Python, DeepFace, OpenCV
 - Built a real-time system that identifies and classifies facial expressions from a live video stream using deep learning inference at interactive frame rates
- **Question Paper Builder — Desktop Application** [Code]
Tools: C++, Qt Framework
 - Built a GUI desktop tool for teachers to compose and format exam question papers, cutting preparation time from several hours to under 15 minutes
 - Packaged and released cross-platform binaries: a Windows .exe installer and a Linux AppImage for portable, dependency-free distribution

HONORS AND AWARDS

- **First Place, Coding Competition**, GCES IT Expo 2024, Gandaki College of Engineering and Science 2024
 - Secured first place in coding competition, demonstrating advanced problem-solving and programming skills
- **Emerging Basketball Player Award**, GCES Basketball Team 2021
 - Recognized as the emerging player on the GCES Basketball Team
- **First Place, [GCES Sports meet]**, Girls’ Cricket Team (Coach) 2023
 - Coached the GCES girls’ cricket team to a first-place finish
- **Project Showcase**, GCES IT Expo, Gandaki College of Engineering and Science 2023
 - Presented an AI-based Emotion Detection System, demonstrating real-time facial emotion recognition to a broad audience

PROFESSIONAL EXPERIENCE

- **Full Stack Developer, Giga Infotech Pvt. Ltd.** July 2025 – June 2026
Kathmandu, Nepal
 - Developed and integrated AI-powered features including text analysis, document processing, and LLM-based automation workflows to enhance user productivity and reduce manual effort
 - Worked directly with clients to gather requirements, translate business needs into technical solutions, and deliver projects on schedule
 - Managed and debugged existing codebases, identifying and resolving issues while maintaining system functionality and reliability
- **Django Developer, DÈs Vu Technologies Pvt. Ltd.** May 2024 – June 2025
Pokhara, Nepal
 - Developed and maintained scalable web applications using Django and Django REST Framework, implementing robust backend architectures and RESTful APIs for production systems
 - Designed and optimized PostgreSQL database schemas, improving query performance and ensuring reliable data management across multiple client projects
 - Collaborated with frontend developers to integrate APIs, troubleshoot issues, and deliver feature enhancements while maintaining code quality through testing and code reviews

TECHNICAL SKILLS

- **Machine Learning / AI:** Deep Learning (PyTorch), Computer Vision, Natural Language Processing, Scikit-Learn, LLMs, RAG-based Information Retrieval, NumPy, Pandas
- **Programming Languages:** Python, C++, JavaScript, Go
- **Backend Development:** FastAPI, Django, Redis, PostgreSQL
- **Frontend Development:** React.js, React Redux, Bootstrap, Tailwind CSS, Next.js
- **Tools & DevOps:** Git/GitHub, Docker, GitLab, Bash, GitHub Actions

CERTIFICATIONS

- **Inclusive Open Source Community Orientation** — The Linux Foundation (Online)